

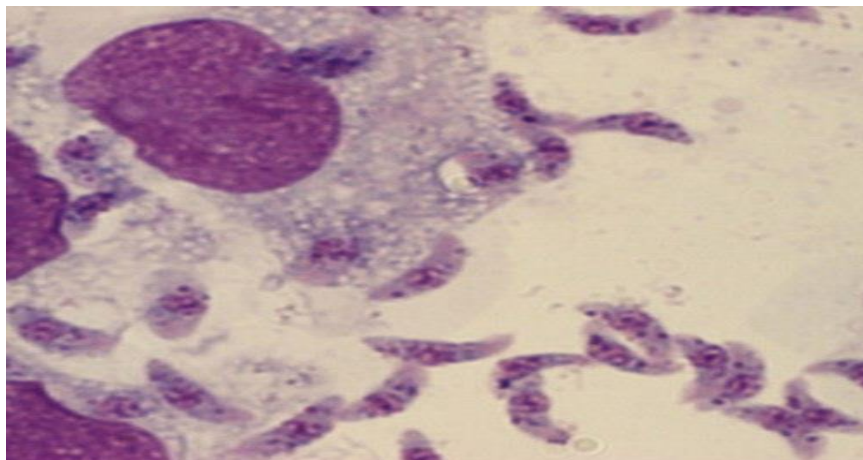
Toxoplasmosis

1

Etiology

Toxoplasma gondii

- An obligate intracellular parasite of the coccidia subclass displaying typical sexual and asexual life cycles

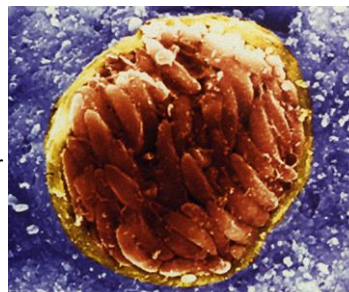


2

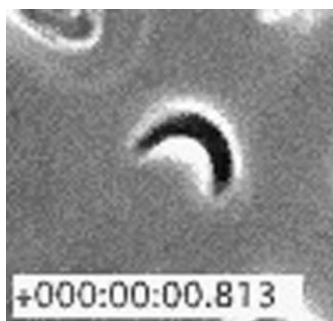
- There are three forms of *Toxoplasma gondii*



Tachyzoite – Rapidly reproducing form



Bradyzoite – A slower reproducing form, contained in tissue

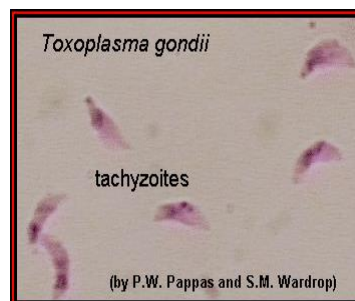


Sporozoite – Contained in oocysts

3

Tachyzoite

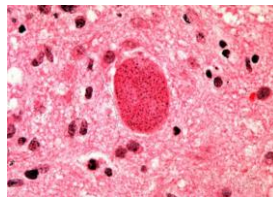
- Tachyzoites invade cells in the body in order to multiply. They invade a cell, multiply themselves, destroy the cell, and this releases more tachyzoites to infect other cells.
- Tachyzoites are seen in many tissues and organs throughout an infected body during the acute phase of the disease.
- The **acute phase** is also called the **extraintestinal phase** because it affects cells outside the intestines.
- Only cats show the intestinal phase of the infection.



4

Bradyzoites

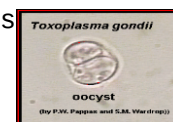
- Two or three weeks after the first infection, the *Toxoplasma* microorganism begins to divide more slowly and a protective membrane forms around the parasite cells.
- The cysts containing the parasite cells are called zoitocysts and the cells inside the cysts are called bradyzoites.
- The tissue cysts are formed primarily in brain, eye, heart muscle, and skeletal muscle.
- Bradyzoites persist in tissue for many years, possibly for the life of the host.



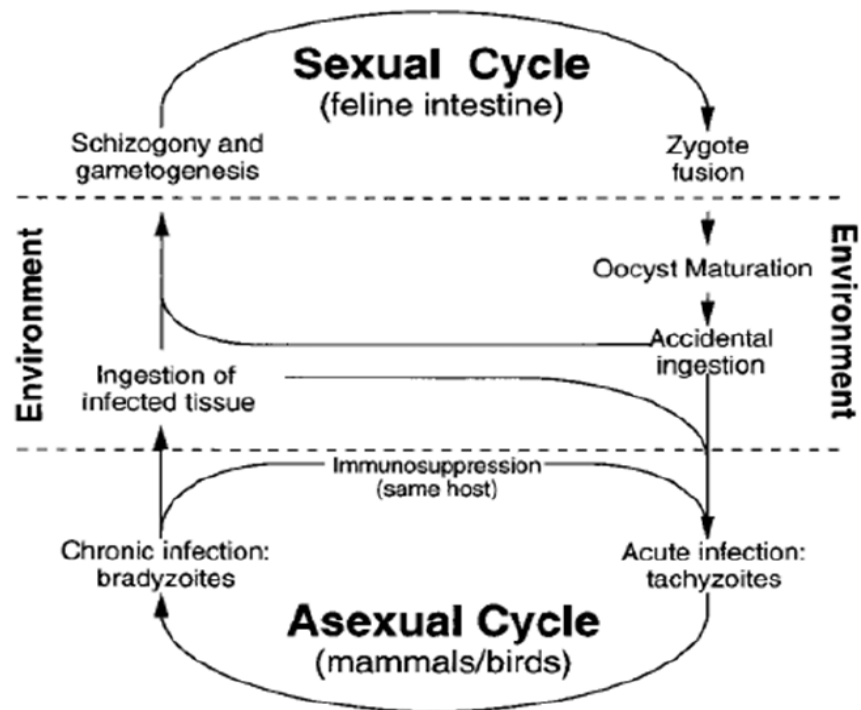
5

Oocysts

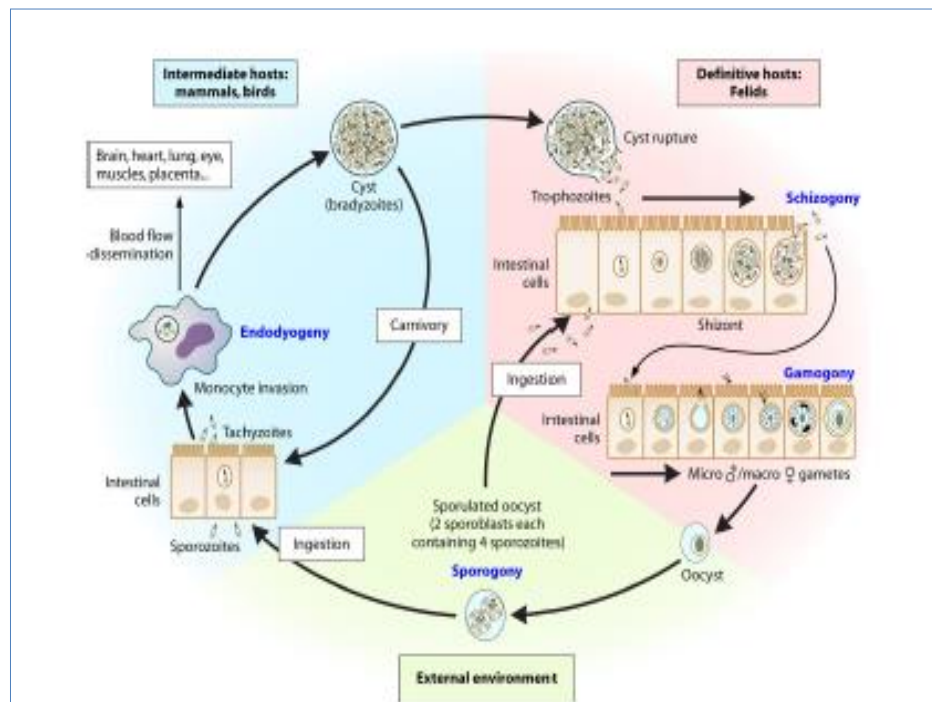
- In cats, the *Toxoplasma* parasites infect the lining of the small intestine where they reproduce asexually.
- After a few days of rapid reproduction the cells transform into a sexual form, combine, and become enclosed in a cyst called an oocyst.
- Oocysts contain the sporozoite form of the *Toxoplasma* parasite.
- Oocysts are found in both wild and domestic cats but not in any other animals or birds.
- Animals become infected by ingesting food contaminated with oocysts from feces or with tissue cysts (bradyzoite) from the flesh of infected animals.
- Infected cats shed millions of oocysts in their feces during the two-week period of the intestinal phase of the infection
- These oocysts can survive in soil for several months and are the major source of infection.
- Once swallowed, the oocyst burst in the intestines and spreads to the rest of the body through the bloodstream.



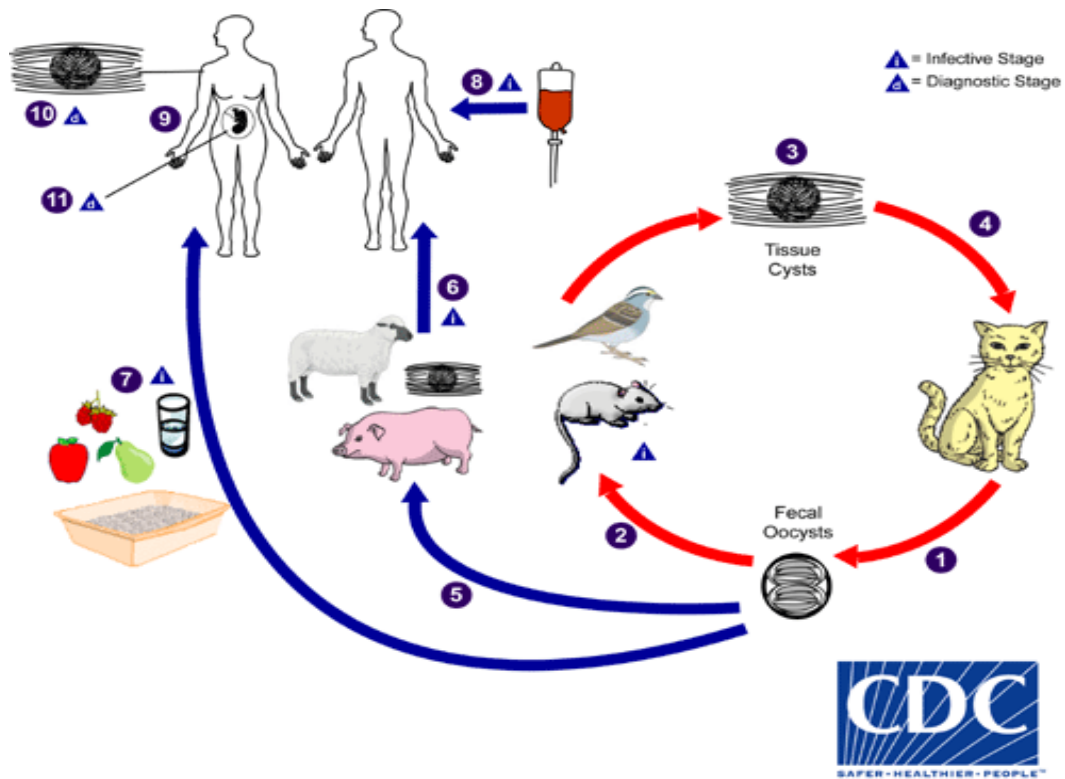
6



7



8



9

The Disease: toxoplasmosis

Immunocompetent individuals/Acquired

Immune compromised individuals/Reactivated

➤ Opportunistic infection

Congenital toxoplasmosis/Pregnancy associated

10

Acquired toxoplasmosis

➤ Immunocompetent children who acquire infection postnatally generally do not have clinically recognizable symptoms.

Symptomatic infection is characterized: lymphadenopathy, fever, and hepatosplenomegaly

➤ Disseminated infection, including myocarditis, pneumonia, and CNS toxoplasmosis, is more common among immuno-compromised persons

➤ Fever, stiff neck, myalgia, arthralgia, maculopapular rash that spares the palms and soles, localized or generalized lymphadenopathy, hepatomegaly, hepatitis, reactive lymphocytosis, meningitis, brain abscess, encephalitis, confusion, malaise, pneumonia, polymyositis, pericarditis, pericardial effusion, and myocarditis.

➤ Chorioretinitis, usually unilateral, occurs in approximately 1% . The most common manifestation is enlargement of 1 or a few cervical lymph nodes.

11

Congenital toxoplasmosis

➤ Congenital infection results from primary acquired maternal infection during gestation.

➤ The frequency of vertical transmission and the severity of fetal damage depend on the stage of pregnancy when maternal infection occurs.

➤ The placenta plays a main role in the process, as it is both a natural barrier which is supposed to protect the fetus and a target tissue for parasite multiplication. In fact, the placental barrier is more efficient at the beginning of gestation, but becomes more permeable throughout pregnancy, allowing parasite transmission

Transmission efficiency:

10% of cases during the first trimester

30% of cases in the second trimester

60 to 70% of cases in the third trimester

➤ The severity of fetal infection is inversely correlated, since neonates are asymptomatic in more than 80% of cases when infected during the third trimester of gestation. When transplacental transmission occurs during the first trimester, often leads to severe abnormalities or to abortion.

12

Early pregnancy infection outcomes

- Major sequelae include mental retardation, seizures, microcephalus, hydrocephalus, deafness, and psychomotor deficiency
- Eye lesions are also more severe in early pregnancy, where microphthalmia, cataract, increased intraocular pressure, strabismus, optic neuritis, and retinal necrosis can be observed, as can uveitis and retinochoroiditis, possibly leading to blindness if retinal lesions affect the macula

13

Congenital Toxoplasmosis

Congenital infection may present as a mild or severe neonatal disease

Main signs: premature birth, small size for gestational age, retinal scars, persistent jaundice, mild thrombocytopenia, CSF pleocytosis

More severe signs:

Psychomotor retardation and resulting low IQ
Seizures / Blindness, sometimes deafness

Severe disease and death

Characteristic triad: chorioretinitis, hydrocephalus, and cerebral calcifications

14

How is toxoplasmosis acquired?

Transmission

Via Oocysts (containing sporozoites)

Fecal-oral (ingestion of contaminated food items, water, fruits, etc...)

Via Cysts (containing bradyzoites)

Eating raw or contaminated meat/prey (through cysts)

Via Tachyzoites

Transplacental (vertical transmission)

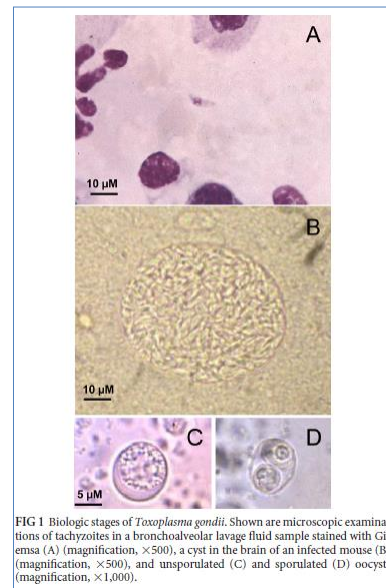
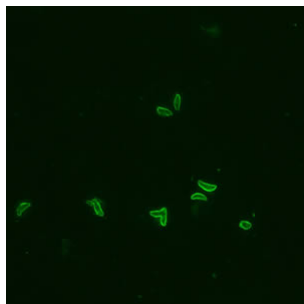
Unpasteurized milk consumption (risk factor)

Blood transfusion

15

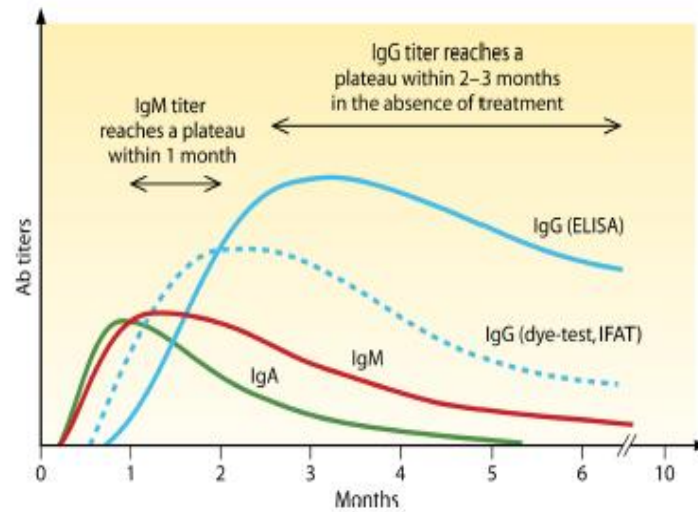
Diagnosis

- History
- Serology testing – measuring the IgM & IgG antibodies
- Sabin-Feldman Dye Test (DT)
- Differential agglutination test (also known as the "AC/HS test")
- Immunofluorescence



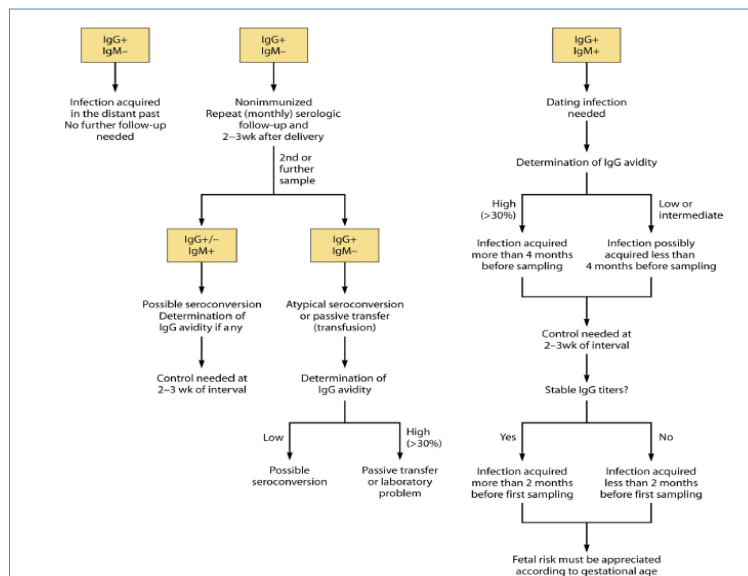
16

Kinetics of Antibody production



17

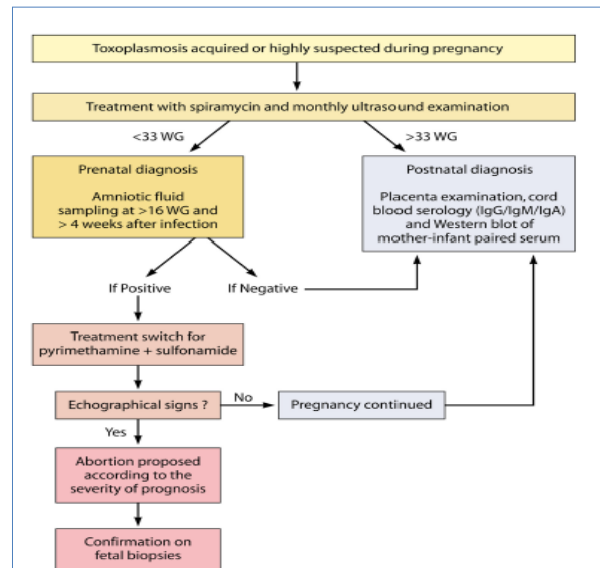
Interpretation of *T. gondii* serologic tests during pregnancy



18

Management of pregnancy with confirmed or highly suspected gestational toxoplasmosis

[WG, weeks of gestation]



19

Prognosis

- For individuals with weakened immune systems, the prognosis is not good. The disease has to be kept down with medicine, and complications can occur depending on what tissues of the body are being affected.
- If toxoplasmosis is affecting the eye, the body will try and attack the foreign invader, which can lead to blindness.
- Immune-compromised hosts infected with toxoplasmosis may have encephalitis, pneumonitis and myocarditis as manifestations of the infection.
- These infections are usually fatal if not recognized and treated.

20

TREATMENT

Acquired Toxoplasmosis and lymphadenopathy

- Do not require specific treatment unless they have severe and persistent symptoms or evidence of damage to vital organs.
- If such signs and symptoms occur, treatment with pyrimethamine, sulfadiazine, and leukovorin should be initiated

21

TREATMENT

Congenital Toxoplasmosis:

- All newborns infected with *T. gondii* should be treated whether or not they have clinical manifestations of the infection because treatment may be effective in interrupting acute disease that damages vital organs.
- Infants should be treated for 1 yr with pyrimethamine, sulfadiazine and leukovorin

22

ALTERNATIVE TREATMENT

Azithromycin: 900-1,200 mg/kg/day with pyrimethamine and leucovorin (B II), but not evaluated in children

Clindamycin with pyrimethamine leucovorin

Adults – atovaquone: 1,500 mg orally BID plus pyrimethamine and leucovorin (C III), but not evaluated in children

Limited use of corticosteroids as adjuvant therapy with CNS disease

23

Prevention

- Avoid eating raw meat, unwashed fruit and vegetables
- Wash hands after dealing with raw meat, gardening, and changing litterboxes.
- Change litterboxes daily (the oocytes shed in the feces don't become infectious for 1-5 days)
- Don't feed animals raw meat.
- If pregnant, wear gloves when changing litterboxes, and keep cats indoors to insure they don't hunt and eat contaminated prey/meat.
- There is only one vaccine available. It is exclusively for sheep and it is only approved for use in Great Britain

24

Prevention

Wash cutting boards, sinks, knives and other utensils that touched raw meat using soap and hot water.